

Zaineb MESSAOUDI

Software Engineering Student · Full-Stack & AI

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VIEW ONLINE

Software Engineering Student · 4 full-stack & AI internships · Fintech, healthcare, edtech
Stack: React, NestJS, Python, FastAPI, MongoDB

PROFESSIONAL SUMMARY

Third-year Software Engineering student at ESPRIT (specializing in Web & Internet Technologies) with four internships in full-stack development across fintech, healthcare, and edtech. Proficient in interface design, backend architecture, deployment, and AI integration. Seeking a 6-month or longer final-year internship starting January in Software Engineering, Web Development, or Applied AI/ML.

WORK EXPERIENCE

Full Stack Developer Intern

06/2026 – 08/2026

QCMed · Remote

- Designed and delivered full-stack features (React + NestJS) for quiz and challenge modules: added detailed question attributes, granular resident progress tracking
- Developed a REST API in Node.js exposing answer statistics by question with MongoDB aggregations, real-time synchronized admin dashboard via WebSocket
- Optimized challenge scoring engine for partial-credit answer validation, reducing manual review time and strengthening user feedback relevance
- Integrated notification workflows on AWS SES for residents and administrators

Web Developer Intern

07/2025 – 08/2025

Esprit Tech · Ariana, Tunis

- Delivered HIKMALEARN, a full-stack accessibility platform for students with disabilities, in **2 months**: designed ReactJS frontend and Django REST backend, deployed to production and passed WCAG **2.1** audit
- Integrated four critical accessibility features: automated video captioning, screen reader support, keyboard navigation, AI-generated image descriptions
- Cross-functional collaboration on a **6**-person developer team: systematic code reviews, short agile cycles, two-week sprint delivery

Web Developer Intern

02/2024 – 06/2024

Banque Centrale de Tunisie (BCT) · Tunis

- Designed and deployed a secure web platform replacing a manual Excel-based inter-departmental event coordination process, reducing collaboration cycle by **40%**
- Implemented structured calendar management, logistics, and archives with granular access control (authentication, role-based permissions) and complete audit trail
- Backend stack: Spring Boot, Java, Thymeleaf, SQL; UX/UI prototyping in Figma; delivery in **2**-week agile sprints

Odoo Developer Intern

07/2022 – 08/2022

I-Way · Manar 1, Tunis

- Designed and deployed an Odoo module (Python, JavaScript) automating patient appointment booking workflow, eliminating manual processing, reducing per-file processing time by **60%**
- Optimized application performance via PostgreSQL query tuning and application-level caching, stabilizing **P95** latency under **500ms** at nominal load
- Integrated module with existing business processes, cross-team functional validation, and active support during production deployment

PROJECTS

TalentLens

01/2026 – 05/2026

Lead Machine Learning

- HR intelligence platform developed within a **5**-person team, leading the Career Machine Learning module
 - Developed models for salary prediction and career trajectory classification, achieving $R^2 = 0.88$ for salary regression and AUC = **0.75** for career classification
 - Applied SHAP explainability to improve model interpretability and provide transparent, actionable insights
 - Contributed to anomaly detection, employee segmentation, and job recommendation modules
 - Stack: Python, XGBoost, LightGBM, Scikit-learn, SHAP, Pandas, React
- Python · XGBoost · LightGBM · Scikit-learn · SHAP · Pandas · React

Oralis

01/2026 – 05/2026

Full Stack Developer

- AI-powered full-stack platform for evaluating oral presentations within a **5**-person team, combining speech-to-text conversion, filler word detection, pause analysis, and pronunciation scoring
- Implemented JWT authentication and role-based access control (RBAC) to secure platform access and user permissions
- Containerized the application with Docker and contributed to a CI/CD pipeline for streamlined deployment
- Stack: React, NestJS, FastAPI, Python, MongoDB, Docker

React · NestJS · FastAPI · Python · MongoDB · Docker · JWT · RBAC

Pregnancy & Maternity Tracking Platform

01/2025 – 05/2025

Full Stack Developer

- Designed and delivered a multi-platform pregnancy and maternity tracking application covering Web, Mobile, and Desktop within a **6**-person team
- Built web, mobile, and desktop clients while maintaining consistent data synchronization across all three platforms
- Integrated Firebase services and managed SQL-based data persistence
- Stack: Symfony, PHP, FlutterFlow, Firebase, JavaFX, SQL

Symfony · PHP · FlutterFlow · Firebase · JavaFX · SQL

SKILLS

LANGUAGES

Java · TypeScript · C / C++ · PHP · JavaScript · Python

BACKEND

Symfony · REST APIs · FastAPI · NestJS · Node.js · Django · Spring Boot · Firebase

DEVOPS & CLOUD

Git/GitHub · Jenkins · Kubernetes · Docker · DevOps

METHODOLOGIES

Scrum · Kanban · Agile

TOOLS

Figma · Adobe Creative Suite · Jira

PROFESSIONAL SKILLS

FlutterFlow · Multitasking & Time Management

FRONTEND

JavaFX · React · Angular

DATABASES

MongoDB · MySQL · PostgreSQL

DATA & AI

Machine Learning · NLP · AI / Artificial Intelligence

SOFT SKILLS

Team Leadership

GRAPHIC DESIGN

Graphic Design

EDUCATION

Engineering Degree – Software Engineering (TWIN - Web & Internet Technologies)

10/2027

ESPRIT School of Engineering

Bachelor's – Computer Science & Multimedia

06/2024

ISAMM

LANGUAGES

Arabic · **Native** · French · **Fluent** · English · **Intermediate**

CERTIFICATIONS

MongoDB CRUD Operations

12/2025

MongoDB

Prompt Engineering and Generative AI

06/2025

Coursera (ODC Free Training Program)

Microsoft Azure Fundamentals (AZ-900)

07/2024

Microsoft Learn

Full Stack Web Development

07/2023

freeCodeCamp

Machine Learning with Python

06/2023

freeCodeCamp

INTERESTS

Open Source · Artificial Intelligence · Design · Reading